AER1516: Robot Motion Planning

Assignment #3 — Autonomous Frontier-Based Exploration

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1 Coverage Plots

Figure 1 compares coverage versus exploration step for the three strategies (random, nearest, custom) on each of the three provided maps. Table 1 summarises the final metrics and threshold compliance. **All nearest runs exceed their grading thresholds, and custom outperforms nearest on the cave.**

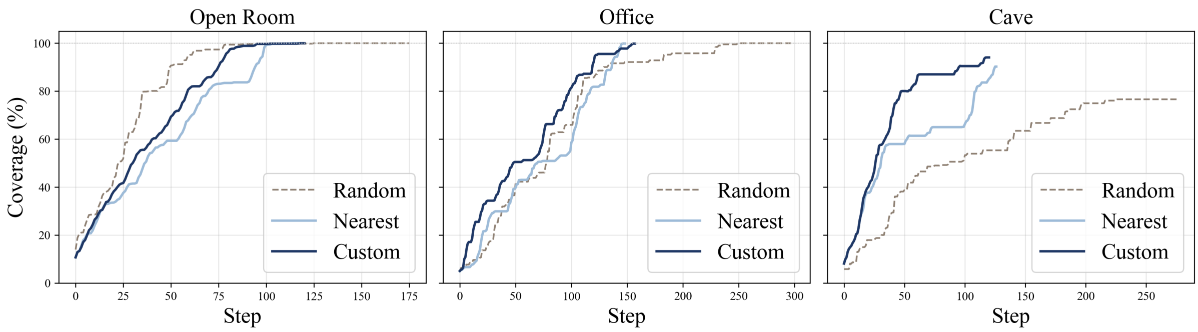

Figure 1: Coverage (%) vs. exploration step.

On the **open room**, all three strategies reach 100% coverage, but random requires 13× the distance of nearest (1707 m vs. 133 m). On the **office map**, nearest and custom both converge above 99% within 150 steps, while random needs 315 steps and over 4 km of travel. On the **cave map**, random stalls at 77%, nearest reaches 90.3% and custom achieves the highest coverage at 94.1% with the shortest distance (141 m).

Table 1: Final exploration metrics (seed = 1024).

Map	Strategy	Coverage (%)	Threshold (%)	Distance (m)	Steps	Pass?
open_room	random	100.0	—	1707	175	—
	nearest	100.0	75	133	119	✓
	custom	100.0	—	146	132	—
office	random	100.0	—	4430	315	—
	nearest	99.9	60	177	150	✓
	custom	99.8	—	185	204	—
cave	random	76.6	—	2980	265	—
	nearest	90.3	55	156	128	✓
	custom	**94.1**	**>90.3**	**141**	**122**	✓

2 Discussion

TL;DR.

Why do certain strategies perform better on certain maps? When frontier values are uniform (open room), pure distance works fine. When dead-ends create low-value traps near the robot (cave), a value-weighted score outperforms greedy nearest by 3.8 pp coverage and 15 m shorter distance.

Where does the greedy nearest-frontier approach struggle? On the cave map, nearest enters nearby dead-end branches first—each yields little new coverage but costs a full round-trip—stalling at 90.3% with 156 m traveled.

What does your custom strategy capture that nearest-frontier misses? Frontier size and local wall density: small, wall-dense frontiers are penalised as likely dead-ends, letting the robot prioritise the main loop and reach 94.1% in 141 m.

Our custom strategy is a dead-end avoidance heuristic that ranks frontiers by

$$\text{score}(f) = \frac{s_f}{c_f \cdot (1 + w_f)}$$

where s_f is frontier size, c_f is path cost, and w_f is the local occupied-cell count in a 7×7 window. Higher scores favour large, close, open-space frontiers and penalise small, wall-dense ones. Theoretically, **our custom strategy** captures two signals that nearest-frontier ignores: frontier size and local openness, so it can bypass nearby traps in favour of more productive frontiers.

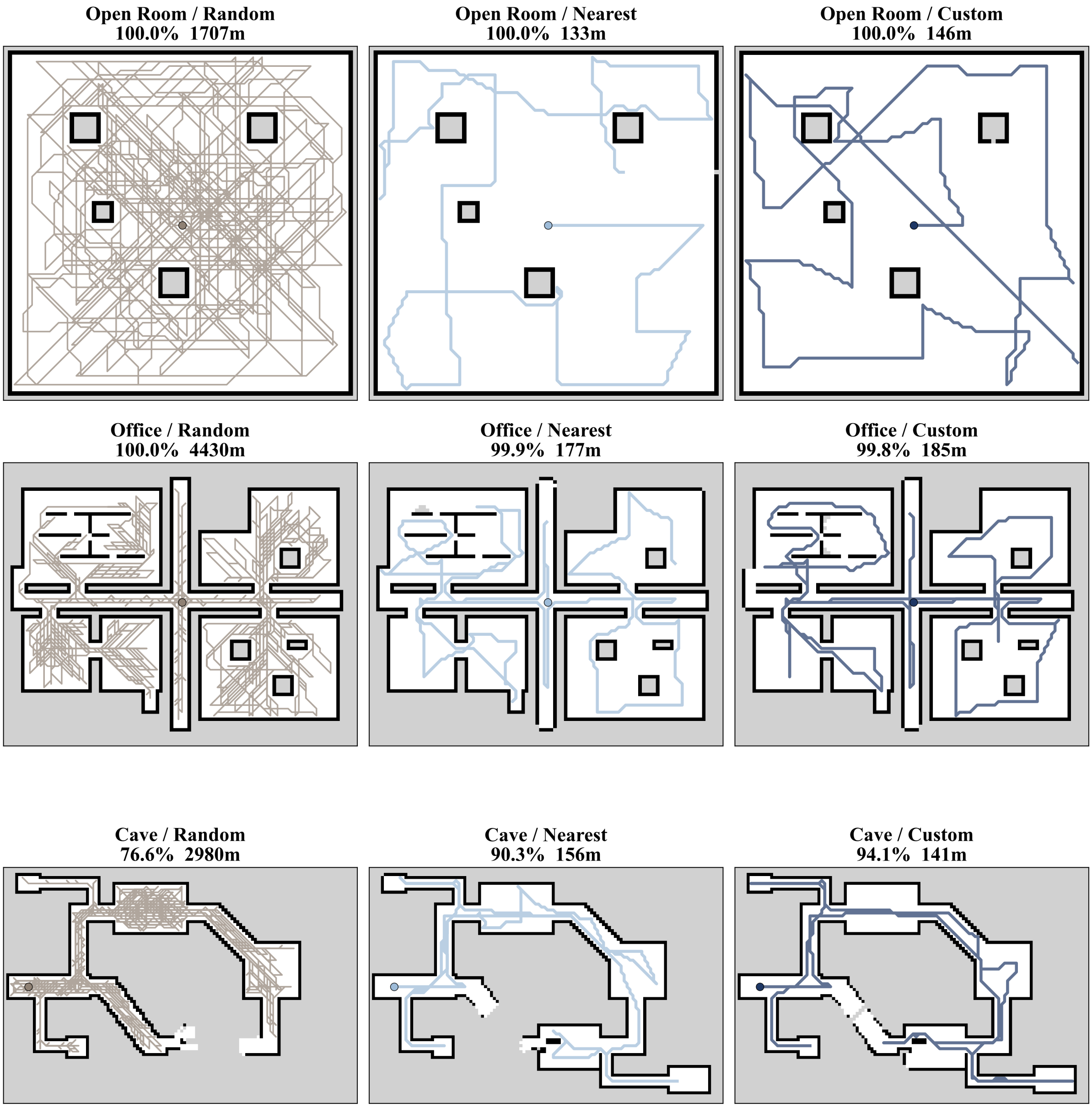

Figure 2: Exploration trajectories for all strategy–map combinations.

On the **open room**, frontier sizes and wall densities are nearly uniform, so the score approximates pure inverse distance; both nearest and custom strategies reach 100% coverage with similar distance (133 m vs. 146 m).

On the **office**, the wall-density term is an imperfect proxy—useful doorway frontiers are also locally wall-dense—so the two strategies remain very close and the benefit of the custom score is small (99.9% / 177 m vs. 99.8% / 185 m).

On the **cave**, the heuristic is well aligned with exploration value: dead-end branches have small, wall-dense frontiers, while the main loop tends to produce larger and more open frontiers. Custom routes through the main loop first (Figure 2, bottom-right), achieving 94.1% in 141 m versus nearest's 90.3% in 156 m—higher coverage and shorter distance.